

SkiPrepCoach

System Design Specification

Version 0.1 (Living Design Document)

1. Vision

Mission

SkiPrepCoach is an adaptive coaching system whose purpose is to help people become healthier, more capable skiers throughout their lives.

Unlike traditional fitness apps, SkiPrepCoach does not prescribe a fixed workout calendar. Instead, it continuously evaluates the athlete's current capabilities, recovery, readiness, goals, and training history to recommend the highest-value training session for today.

The long-term objective is not simply to ski better next season, but to remain a strong, resilient skier for decades.

2. Design Philosophy

Primary Goals

1. Maximize skiing capability.
 2. Minimize injury risk.
 3. Maximize long-term health.
 4. Build lifelong athleticism.
 5. Make training enjoyable.
 6. Encourage consistency over intensity.
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Guiding Principles

P1 Health before performance

Today's workout should never compromise tomorrow's ability to train.

P2 Capability over appearance

The system measures what the athlete can do—not how they look.

P3 Progress through demonstrated ability

Levels advance only through successful gate tests.

Time alone never advances a level.

P4 Minimum Effective Dose

Provide the smallest stimulus that produces adaptation.

Avoid unnecessary fatigue.

P5 Recovery is training

Recovery is part of the training plan.

P6 Measure everything possible

If something can be measured objectively, it should be.

P7 Never waste a Green day

Healthy athletes should continue progressing.

P8 Respect recovery

Every tissue has its own recovery timeline.

P9 Build complete athletes

Although skiing emphasizes the lower body, the system trains the entire body.

Upper body strength, grip, mobility, reaction, balance, and cardiovascular fitness all contribute to skiing and injury resilience.

P10 Explain every recommendation

Every recommendation should answer:

Why today?

Why this exercise?

Why not something harder?

3. Training Philosophy

Training progresses through:

Safety

↓

Movement Quality

↓

Consistency

↓

Capacity

↓

Strength

↓

Power

↓

Skill

↓

Performance

The system never skips stages.

4. Capability Model

The athlete is represented as a collection of capabilities.

Each capability develops independently.

Examples:

 Lower Body Strength

 Posterior Chain

 Upper Body Strength

 Core Stability

 Foot Strength

 Knee Capacity

 Bone Health

 Balance

 Reaction Speed

 Coordination

 Cardio Capacity

 Aerobic Endurance

 Stamina

 Ski Endurance

 Mobility

 Flexibility

 Recovery

 Fall Resilience

 Hiking Capacity

 Rollerblade Skill

 Court Agility

Each capability has:

- Current score
- Trend
- Recovery estimate
- Historical progress

The limiting capability receives priority.

5. Levels

The system contains progressive levels.

Each level contains:

Mission

Exercises

Gate tests

Advancement rules

Coach notes

Reference section

A level does NOT represent a fixed workout schedule.

It represents the maximum safe capability currently available.

6. Gates

Levels advance only after successful gate completion.

Gate tests are objective.

Examples:

Wall Sit

Single-leg balance

Rollerblade distance

Single-leg squat

Hiking duration

Strength targets

Pain response

Next-morning recovery

Gate success requires:

Required Green days

Required successful gate attempts

No significant adverse recovery

7. Exercise Encyclopedia

Every exercise is defined once.

Exercise Specification includes:

ID

Icon

Name

Description

Purpose

Instructions

Equipment

Load

Tempo

Rest

Target RPE

Pain limits

Progressions

Regressions

Capabilities trained

Recovery class

Safety notes

Estimated duration

Common mistakes

Coaching cues

8. Workout Prescription

Exercises are prescribed using compact notation.

Example:

 GSQ 20kg 3×10 3-1-2 R60 RPE6

Meaning:

Goblet Squat

20 kg

3 sets

10 reps

3 sec down

1 sec pause

2 sec up

60 sec rest

Target effort 6/10

Every parameter is measurable.

No instruction should be ambiguous.

9. Recovery Model

Recovery is capability-specific.

Heavy squats may require:

72 hours for heavy lower-body strength

while allowing:

Upper body

Mobility

Balance

Reaction

to continue progressing.

Recovery classes:

Daily

Light

Moderate

Heavy Strength

Maximum Strength

Plyometric

Long Endurance

Recovery estimates improve over time using athlete history.

10. Readiness Model

Every day begins with a readiness check.

Inputs include:

Pain

Swelling

Morning stiffness

Sleep

Energy

Resting HR

HRV status (optional)

Body Battery (optional)

Training Readiness (optional)

Readiness results:

 Green

 Yellow

 Red

Green:

Progress normally.

Yellow:

Reduce load or hold level.

Red:

Recovery session.

11. Sensor Layer

The coaching engine never depends on hardware.

All sensors feed a common interface.

Possible providers:

Manual

Garmin

Apple Health

Health Connect

Whoop

Oura

CSV Import

MVP uses manual entry.

Sensors provide evidence—not decisions.

12. Evidence Layer

Sensor data is transformed into evidence.

Examples:

Resting HR elevated

↓

Recovery likely reduced

Poor sleep

↓

Recovery likely reduced

Pain 4/10

↓

High confidence training restriction

Every evidence item includes a confidence level.

The Coach reasons from evidence.

13. The Coach

The Coach is the decision engine.

It answers one question:

What is the highest-value training session today?

The Coach considers:

Readiness

Recovery

Capabilities

History

Goals

Season

Equipment

Available time

Gate progress

Preferences

It does NOT follow a fixed calendar.

14. Council of Coaches

Decision making is divided into specialized modules.

Safety Coach

Has veto authority.

Recovery Coach

Determines trainable systems.

Performance Coach

Selects today's objective.

Technique Coach

Ensures movement quality.

Motivation Coach

Maintains engagement.

Future coaches may be added.

15. Decision Pipeline

1. Is training safe?
 2. What has recovered?
 3. What capabilities are limiting?
 4. What capabilities are most important this season?
 5. Is a gate approaching?
 6. Select today's objective.
 7. Assemble today's workout.
 8. Explain why.
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16. Workout Scheduler

The scheduler assembles today's workout.

Inputs:

Recovery

Readiness

Capabilities

Exercise history

Weekly volume

Recovery windows

Outputs:

Today's workout

Reason codes

Estimated duration

Expected RPE

Expected adaptation

17. Workout Player

The athlete performs one task at a time.

The player controls:

Timers

Tempo

Rep counting

Rest

Pain recording

Completion

The player automatically saves progress after every action.

Closing the browser never loses progress.

18. Session Flow

Morning Check-in

↓

Coach Decision

↓

Daily Briefing

↓

Workout Player



Reflection



Summary



Recovery Prediction



Tomorrow Reminder

19. User Experience

The app should feel like a personal coach.

The athlete never builds workouts.

The athlete simply answers questions and begins today's mission.

The app should explain:

What

Why

How

What improved

What remains

20. Data Philosophy

The app is deterministic.

The same inputs always produce the same recommendation.

AI is optional.

AI may:

Explain

Teach

Motivate

Summarize

AI never controls progression or safety.

21. Local-First Architecture

Entire application state is stored locally.

Local Storage contains:

Program

Capabilities

Exercises

Sessions

History

Current workout

Settings

Recovery

Messages

Future versions may synchronize this state to cloud storage without changing the coaching engine.

22. JSON-Driven Content

All content is data.

Examples:

Exercises

Levels

Capabilities

Gate tests

Recovery classes

Coach messages

The application logic should rarely require changes when content evolves.

23. Message System

The Coach uses deterministic templates.

Example:

Reason Code:

heavy_lower_recovering

Message:

"Heavy lower-body work is unavailable today because your previous session is still within its recovery window."

Messages are assembled from reusable components.

No LLM is required.

24. Biometrics

Initially entered manually.

Supported metrics:

Resting Heart Rate

HRV Status

Sleep Score

Body Battery

Training Readiness

These influence—but never override—pain and safety signals.

25. Long-Term Vision

SkiPrepCoach is intended to become a comprehensive adaptive coaching platform.

Future capabilities may include:

Additional ski campaigns

Healthy Aging

Running

Cycling

Hiking

Strength

Post-injury rehabilitation

The coaching engine should remain reusable across all campaigns.

26. MVP Scope

The first version will include:

- ✓ Local-first architecture
- ✓ Manual biometric entry
- ✓ Exercise encyclopedia
- ✓ Level progression
- ✓ Daily readiness
- ✓ Workout player with timers
- ✓ Gate testing
- ✓ Recovery model
- ✓ Capability tracking
- ✓ Deterministic coach
- ✓ JSON-based content

No cloud services are required.

No wearable integration is required.

No AI is required.

27. Future Enhancements

Potential future additions include:

- Automatic wearable synchronization (Garmin, Apple Health, etc.)
- Camera-based form analysis
- Adaptive interval timers based on heart rate
- Video demonstrations
- Voice coaching
- Seasonal training plans

- Multiple campaigns (skiing, hiking, longevity, etc.)
 - Multi-device synchronization
 - Community-shared exercise libraries and campaigns
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28. The Prime Directive

Every recommendation made by SkiPrepCoach should satisfy this question:

If the athlete follows this recommendation consistently for the next 20 years, will they become a healthier, more capable, more resilient skier while minimizing unnecessary injuries and continuing to enjoy the process?

If the answer is "no," the recommendation should not be made.

This principle takes precedence over all other design decisions.